Why Algorithms Need the Machine

Locality, Speed, and the 16-Structure Sprint

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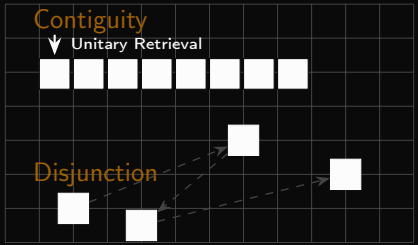
The Spatial Premise



- Modern CPUs operate on a **spatial premise**.
- Contiguous layouts leverage hardware prefetching.
- Pointer-chasing induces serial stalls.
- Memory latency is the **primary bottleneck** of the 21st-century.

The Scale of the Project

16
Data Structural Families

6,641
High-Fidelity Datapoints

722.2B
Memory Elements Analyzed

The Reality: Big O is Only Half the Story

Most students learn algorithms as a count of "steps." Hardware is different.

"Memory Access is like a Coffee Shop vs. a Warehouse"

L1 Cache (Coffee Shop)
Right in front of you. Fast. Automatic.

RAM (The Warehouse)
Slow. Requires a trip. CPU stalls while waiting.

Choosing a data structure that "keeps data close" is the secret to speed.

Meet the Contenders: Beyond the BST

How do we organize data when a simple list isn't enough?

SkipList

The Express Elevator

A list with "shortcut" lanes to skip nodes. Fast as a Tree, but simpler.

vEB Tree

Digital Fractals

Uses recursion to divide data into clusters. Theoretically the fastest.

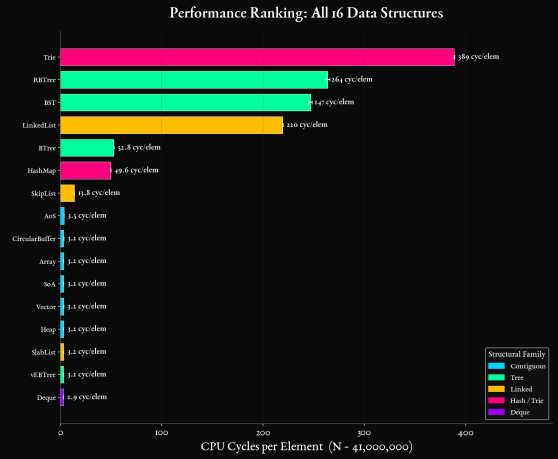
B-Tree

The Librarian

Stores hundreds of items per node. Keeps the CPU happy by minimizing trips.

Measured all 16 contenders across 5 orders of magnitude.

Who Wins the Sprint? (N = 41 Million)

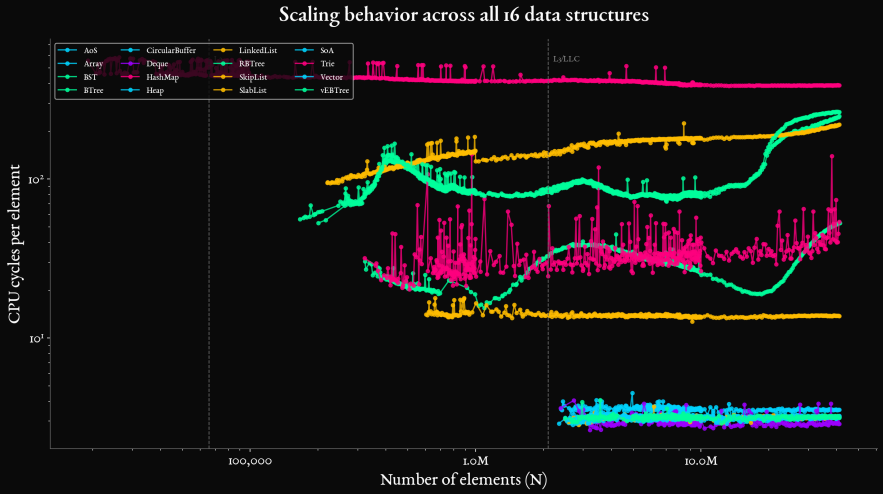


The Latency Gap

Arrays win by 40x.

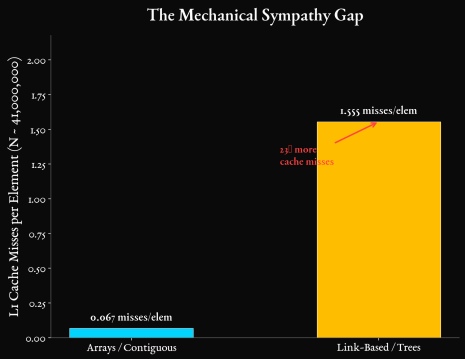
Linked structures stay in "the warehouse" while the CPU waits.

Size Matters: The Cache Cliff



Notice the "step up" in cycles as data outgrows the CPU's internal memory.

The Secret Multiplier: Locality



> *"Locality is like Shopping in one aisle vs. running across the store."*

Pointer Tax: Lists force the CPU to find a new address for every single element.

Practical Lessons

Algorithms are Math, but Performance is Physics.

I. **Contiguity is King:** Keep your data together in memory.
II. **Complexity $\neq$ Speed:** A smart algorithm can be slow if it's "messy."
III. **Hardware Harmony:** Think about the machine, not just the code.

Any Questions?

Thank You!

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github.com/SheikhMohsin9311/Performance-and-Data-Representation